

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2108

Urbana Perkins Road 138 kV
October 27, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2108. The Interconnection Customer proposes a 107.1 MW solar photovoltaic facility interconnecting at the Urbana Perkins Road 138 kV substation in the Pennsylvania area. The request was received on June 10, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 635 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$63,879,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Heardmont to Urbana Perkins Rd Substation 138kV
Contingency: White Swan to Gisler 138kV Line
Loading: 151.4% (Rating: 165 MVA)
DFAX: 100.0%
Pre-project loading: 86.6%
2. Facility: Nueva Guinea 3405 to Heardmont 138kV L
Contingency: White Swan to Gisler 138kV
Loading: 139.4% (Rating: 175 MVA)
DFAX: 100.0%
Pre-project loading: 78.0%
3. Facility: Clear Springs 345/138 kV Auto
Contingency: Wgallatin to Walnut 345kV
Loading: 129.0% (Rating: 480 MVA)
DFAX: 5.2%
Pre-project loading: 127.9%
4. Facility: White Swan to S Mtn 138kV
Contingency: Esquon Substation to Kooskia 138kV
Loading: 126.9% (Rating: 280 MVA)
DFAX: 9.3%
Pre-project loading: 123.4%
5. Facility: Nueva Guinea 3405 to Westside 138kV
Contingency: White Swan to Gisler 138kV
Loading: 120.5% (Rating: 196 MVA)
DFAX: 100.0%
Pre-project loading: 65.8%

6. Facility: Polk St to Van Wagner 138kV
Contingency: West Gallatin to Walnut 345kV
Loading: 119.7% (Rating: 445 MVA)
DFAX: 7.5%
Pre-project loading: 117.9%
7. Facility: Verde Mar 2605 to Croton Falls 138kV
Contingency: W Gallatin to Walnut 345kV
Loading: 116.9% (Rating: 512 MVA)
DFAX: 5.7%
Pre-project loading: 115.7%
8. Facility: Wigwam to Ontelaunee Energy Center 345kV
Contingency: Westwood Generation Llc to Morganville 345kV L
Loading: 112.5% (Rating: 1,307 MVA)
DFAX: 10.8%
Pre-project loading: 111.6%
9. Facility: Railhead to Polk St Substation 138kV
Contingency: Walnut to W Gallatin Substation 345kV
Loading: 111.8% (Rating: 489 MVA)
DFAX: 7.5%
Pre-project loading: 110.2%
10. Facility: Railhead to S Mtn 138kV
Contingency: Esquon to Kooskia 138kV
Loading: 111.4% (Rating: 312 MVA)
DFAX: 9.3%
Pre-project loading: 108.2%
11. Facility: Gisler to White Swan 138kV
Contingency: Heardmont to Urbana Perkins Rd 138kV L
Loading: 104.8% (Rating: 239 MVA)
DFAX: 100.0%
Pre-project loading: 59.9%
12. Facility: Los Angeles Hts to Walden North Generation Sta Substation 138kV
Contingency: Sanborn to Van Wagner 138kV
Loading: 101.6% (Rating: 369 MVA)
DFAX: 5.2%
Pre-project loading: 100.1%
13. Facility: Westside to White Bluffs 138kV Line
Contingency: White Swan to Gisler 138kV
Loading: 101.1% (Rating: 224 MVA)
DFAX: 100.0%
Pre-project loading: 53.2%
14. Facility: Pulpgen to Clear Crossing 138kV line
Contingency: Van Wagner to Sanbrn 138kV
Loading: 100.6% (Rating: 330 MVA)
DFAX: 5.2%
Pre-project loading: 98.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Heardmont to Urbana Perkins Rd Substation 138kV	Rebuild line with higher-capacity conductor	\$3,962,000	100% (\$3,962,000)
Nueva Guinea 3405 to Heardmont 138kV L	Rebuild line with higher-capacity conductor	\$4,744,000	100% (\$4,744,000)
Clear Springs 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$20,565,000	26% (\$5,384,000)
White Swan to S Mtn 138kV L	Rebuild line with higher-capacity conductor	\$9,975,000	100% (\$9,975,000)
Nueva Guinea 3405 to Westside 138kV L	Rebuild line with higher-capacity conductor	\$2,232,000	100% (\$2,232,000)
Polk St to Van Wagner 138kV	Rebuild line with higher-capacity conductor	\$5,340,000	100% (\$5,340,000)
Verde Mar 2605 to Croton Falls 138kV	Rebuild line with higher-capacity conductor	\$9,464,000	100% (\$9,464,000)
Wigwam to Ontelaunee Energy Ctr 345kV	Rebuild line with higher-capacity conductor	\$26,057,000	2% (\$455,000)
Polk St to Railhead 138kV	Rebuild line with higher-capacity conductor	\$4,498,000	100% (\$4,498,000)
S Mtn Substation to Railhead 138kV	Rebuild line with higher-capacity conductor	\$5,447,000	100% (\$5,447,000)
Gisler to White Swan 138kV	Reconductor line	\$1,050,000	100% (\$1,050,000)
Los Angeles Heights to Walden N Generation Station 138kV	Reconductor line	\$1,770,000	100% (\$1,770,000)
White Bluffs to Westside 138kV line	Reconductor line	\$4,316,000	100% (\$4,316,000)
Pulpugen to Clear Crossing 138kV line	Reconductor line	\$5,242,000	100% (\$5,242,000)
	Total allocated to project		\$63,879,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.